



SSW3301-02: SOFTWARE REQUIREMENT ENGINEERING

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SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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TABLE OF CONTENTS

1. INTRODUCTION.....	4
1.1 Purpose.....	4
1.2 Document Conventions.....	4
1.3 Intended Audience and Reading Suggestions.....	4
1.4 Product Scope.....	5
1.5 References.....	5
2. OVERALL DESCRIPTION.....	6
2.1 Product Perspective.....	6
2.2 Product Functions.....	6
2.3 User Classes and Characteristics.....	7
2.4 Operating Environment.....	7
2.5 Design and Implementation Constraints.....	8
2.6 Assumptions and Dependencies.....	8
3. EXTERNAL INTERFACE REQUIREMENTS.....	9
3.1 User Interfaces.....	9
3.2 Hardware Interfaces.....	9
3.3 Software Interfaces.....	10
3.4 Communications Interfaces.....	10
4. SYSTEM FEATURES.....	11
4.1 Module 1: User Authentication and Profile Management.....	11
4.1.1 Description and Priority.....	11
4.1.2 Stimulus/Response Sequences.....	11
4.1.3 Functional Requirements.....	11
4.2 Module 2: Search and AR Indoor Navigation.....	12
4.2.1 Description and Priority.....	12
4.2.2 Stimulus/Response Sequences.....	12
4.2.3 Functional Requirements.....	12
4.3 Module 3: Admin Management Portal.....	13
4.3.1 Description and Priority.....	13
4.3.2 Stimulus/Response Sequences.....	13
4.3.3 Functional Requirements.....	13
5. OTHER NONFUNCTIONAL REQUIREMENTS.....	14
5.1 Performance Requirements.....	14
5.2 Safety and Security Requirements.....	14
5.3 Software Quality Attributes.....	15

1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to define the complete functional, non-functional, and interface requirements for the AR-Enabled Campus Navigation and Lab Locator Application. This document serves as the primary specification for the development team, system testers, and stakeholders to ensure all technical and user expectations are met.

1.2 Document Conventions

This document follows the IEEE 830-1998 standard for SRS formulation. The following conventions are applied:

- **Shall:** Indicates a mandatory requirement that the system must satisfy.
- **Should:** Indicates a desirable requirement but not strictly mandatory.
- **Bold text:** Used to emphasize critical technical terms, system components, or specific headings.

1.3 Intended Audience and Reading Suggestions

This document is intended for:

- **SSW3301 Course Instructors/Evaluators:** To review compliance with software requirement engineering best practices.
- **Project Group Members / Developers:** To serve as a definitive baseline guide for implementing code and system modules.
- **Quality Assurance (QA) / Testers:** To design test cases based on the listed system features and validation points.

1.4 Product Scope

The AR-Enabled Campus Navigation and Lab Locator Application is a mobile-based solution designed to help students, staff, and visitors navigate the faculty campus efficiently. By leveraging Industry Revolution (IR) 4.0 technology—specifically Augmented Reality (AR) and Mobile Technology—the app overlays real-time digital directional cues onto the physical campus environment via the user's smartphone camera.

The software will specifically allow users to:

- Locate lecture rooms, laboratories, administration offices, and facilities around the faculty.
- View step-by-step pedestrian routing using a live AR camera feed view.
- Access real-time lab schedules and room information upon arrival.

1.5 References

1. IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*.
2. Group Project SSW3301 Guidelines SEM 2 2025/2026, Faculty of Computer Science and Information Technology.
3. Google ARCore / Apple ARKit Developer Documentation (for AR baseline implementation).

2. OVERALL DESCRIPTION

2.1 Product Perspective

The AR-Enabled Campus Navigation and Lab Locator Application is a new, standalone mobile application tailored specifically for indoor navigation within the faculty buildings of Universiti Putra Malaysia (UPM). The system relies on mobile hardware components (camera, gyroscope, and accelerometer) combined with indoor positioning technologies (such as Wi-Fi fingerprinting, Bluetooth Beacons, or Visual Anchor points) to track user location inside buildings where GPS signals are unavailable. The mobile app interfaces with a centralized Cloud Backend Database maintained by the System Administrator to retrieve updated floor plans, room descriptions, and lab availability data.

2.2 Product Functions

The core capabilities of the application include:

- User Authentication: Secure login system using UPM institutional IDs for students and staff.
- Search & Find: Search engine to look up specific labs, lecture halls, and meeting rooms.
- AR Indoor Navigation: Real-time camera overlay showing directional arrows and paths leading to the selected destination inside the building.
- Lab Info & Schedule Display: Digital data display (pop-up) showing the current class or lab schedule when a user arrives at a specific lab location.
- Content Management (Admin Portal): A backend web interface for administrators to add new rooms, modify lab details, or update spatial layout coordinates.

2.3 User Classes and Characteristics

The system categorizes users into three distinct classes:

- **Students:** General users who utilize the app to search for classrooms, locate labs for practical sessions, and get AR routing instructions. They require an intuitive and highly responsive UI.
- **Staff (Lecturers/Support):** Users who use the app to locate unfamiliar meeting rooms or specific laboratories. They have similar viewing privileges to students but may have additional access to staff-only facility info.
- **System Administrators:** Tech-savvy personnel responsible for maintaining system data. They use the backend web interface to update spatial markers, map assets, schedules, and user access controls.

2.4 Operating Environment

The system operates across two main environments:

1. **Client-Side Mobile Application:**
 - **Android OS:** Version 10.0 or higher with ARCore support.
 - **iOS:** Version 14.0 or higher with ARKit support.
2. **Server-Side Web & Database:**
 - Hosted on a secure cloud server (e.g., AWS or Firebase).
 - Accessible via modern desktop web browsers (Chrome, Firefox, Safari) for administrative tasks.

2.5 Design and Implementation Constraints

- **Hardware Dependence:** The mobile device must possess a functioning rear camera, gyroscope, and accelerometer. Devices without AR compatibility (ARCore/ARKit) will not run the navigation features.
- **Signal Limitations:** Indoor localization depends strictly on campus Wi-Fi strength or active sensor nodes (Beacons). Weak network coverage inside heavy concrete structures may affect tracking precision.
- **Platform Uniformity:** The user interface must maintain look-and-feel consistency between the Android and iOS versions.

2.6 Assumptions and Dependencies

- **Network Availability:** It is assumed that the UPM campus Wi-Fi network or mobile data is continuously available for fetching live database updates.
- **Building Structural Stability:** The application maps assume building floor layouts remain static. Physical renovations or room swaps require immediate manual updates by the Admin to prevent routing errors.

3. EXTERNAL INTERFACE REQUIREMENTS

3.1 User Interfaces

The application shall provide two separate, responsive user interfaces designed for clear readability and ease of use:

- **Mobile Interface (Students & Staff):** A clean, minimalist layout with a prominent Search Bar on the main dashboard. The navigation interface will feature a Split-Screen View or a Full-Screen AR View that overlays digital directional arrows directly onto the smartphone camera feed. Dark Mode and Light Mode options shall be supported based on device settings.
- **Web Portal Interface (Administrators):** A dashboard interface accessible via desktop browsers. It will feature a sidebar navigation menu containing sections for User Management, Lab Schedule Editor, and Floor Map Coordinates Configuration.

3.2 Hardware Interfaces

The mobile application must interface directly with the following hardware components of the student or staff smartphone:

- **Rear Camera:** Used continuously during navigation to capture the physical indoor environment for AR tracking.
- **Inertial Measurement Unit (IMU):** Integrates data from the device's gyroscope and accelerometer to track the user's physical steps, orientation, and angle changes in real-time.
- **Wi-Fi / Bluetooth Module:** Used to scan local network SSIDs and Bluetooth Beacon signals inside the UPM buildings to triangulate the user's initial indoor coordinate position.

3.3 Software Interfaces

The application shall interface with external software services and database engines via secure APIs:

- UPM Single Sign-On (SSO) API: The application shall connect to the official UPM identity provider system via secure protocols (OAuth 2.0 / SAML) to authenticate student and staff credentials during login.
- Google Firebase Suite:
 - Firebase Authentication: To manage user session tokens after successful UPM SSO verification.
 - Cloud Firestore / Realtime Database: To store and sync indoor map nodes, lab names, current schedules, and room coordinates.
 - Firebase Cloud Storage: To store static visual assets, such as 3D AR arrow models and floor plan images.
- AR Development Frameworks: Google ARCore (for Android deployment) and Apple ARKit (for iOS deployment) to handle computer vision calculations, plane detection, and spatial anchoring.

3.4 Communications Interfaces

- HTTPS Protocol: All API requests between the mobile client, admin portal, and Firebase servers must be encrypted using Transport Layer Security (TLS/SSL) via HTTPS.
- WebSockets: The application shall utilize Firebase's native WebSocket-based connection to push live data updates (e.g., immediate lab room description changes by the admin) to active mobile users without requiring manual app refreshes.

4. SYSTEM FEATURES

4.1 Module 1: User Authentication and Profile Management

4.1.1 Description and Priority

This module handles secure access control to the application, ensuring that only verified UPM students, staff, and authorized administrators can access the system's core features.

- Priority: High (Mandatory for system access and security compliance).

4.1.2 Stimulus/Response Sequences

- Stimulus: The user opens the mobile application and taps the "Login with UPM ID" button.
- Response: The system redirects the user to the secure UPM Single Sign-On (SSO) authentication gateway page.
- Stimulus: The user enters valid UPM credentials and submits the form.
- Response: The system authenticates the user, creates a secure session token via Firebase, and navigates to the main navigation dashboard.

4.1.3 Functional Requirements

- FR-1.1: The mobile application shall redirect all unauthenticated users to the login screen upon startup.
- FR-1.2: The system shall interface with the UPM SSO gateway to validate user credentials using OAuth 2.0 protocols.
- FR-1.3: The system shall retrieve and store basic user profile details (Name, Matric/Staff ID, and User Role) into Firebase upon successful authentication.
- FR-1.4: The application shall allow the user to log out securely, which immediately destroys the local session token.

4.2 Module 2: Search and AR Indoor Navigation

4.2.1 Description and Priority

This is the core module of the application. It allows users to search for specific laboratories or lecture halls inside the faculty and guides them using real-time AR camera overlays.

- Priority: High (Core business feature of the application).

4.2.2 Stimulus/Response Sequences

- Stimulus: The user types a laboratory name (e.g., "Makmal Perisian 1") into the search bar.
- Response: The system displays a dropdown list of matching destinations.
- Stimulus: The user selects the desired laboratory from the list and taps "Start Navigation".
- Response: The system activates the rear camera feed and projects 3D AR arrows on-screen, guiding the user step-by-step from their current indoor location to the target lab.

4.2.3 Functional Requirements

- FR-2.1: The application shall provide a search function that queries the Firebase Cloud Firestore database for lab names, room numbers, and floor levels.
- FR-2.2: The system shall automatically detect the user's initial indoor location using spatial visual anchors or local Wi-Fi/Beacon signals.
- FR-2.3: The application shall trigger the smartphone's rear camera to display the live physical environment when navigation is active.
- FR-2.4: The system shall overlay dynamic 3D directional arrows onto the camera screen using Google ARCore or Apple ARKit frameworks to guide the user.
- FR-2.5: The system shall display a digital information card containing the current lab schedule and availability once the user arrives at the lab door.

4.3 Module 3: Admin Management Portal

4.3.1 Description and Priority

This module provides a web-based dashboard interface for system administrators to perform Create, Read, Update, and Delete (CRUD) operations on faculty rooms, map data, and laboratory class timetables.

- Priority: Medium (Required for data maintenance and operational accuracy).

4.3.2 Stimulus/Response Sequences

- Stimulus: An administrator logs into the web backend and selects the "Lab Management" tab.
- Response: The portal fetches and displays a structured list of all registered rooms and schedules from Firebase.
- Stimulus: The administrator edits a lab schedule and clicks the "Save Changes" button.
- Response: The portal writes the updated schedule data to Firebase, which synchronizes immediately with all active mobile client applications.

4.3.3 Functional Requirements

- FR-3.1: The system shall restrict access to the web portal strictly to users verified with the "Administrator" role flag in Firebase.
- FR-3.2: The portal shall provide forms for the administrator to add, modify, or delete laboratory names, room IDs, and operational status.
- FR-3.3: The system shall allow administrators to upload and assign specific time-slot schedules to individual laboratories.
- FR-3.4: The web portal shall provide tools to update spatial node coordinates and map anchor data to maintain AR tracking alignment.

5. OTHER NONFUNCTIONAL REQUIREMENTS

5.1 Performance Requirements

- PR-1 (Response Time): The application shall load and display the search results within 2 seconds of the user typing a query.
- PR-2 (AR Frame Rate): The AR visual overlay shall maintain a minimum rendering frame rate of 30 frames per second (fps) on standard compatible mobile devices to prevent visual lag or motion sickness.
- PR-3 (Synchronization Speed): Any update made by the Administrator via the web portal shall synchronize with the Firebase database and update active user mobile clients within 3 seconds under a stable network connection.
- PR-4 (Localization Accuracy): The indoor positioning engine shall determine the user's location with an accuracy error margin of no more than 1.5 meters inside UPM buildings.

5.2 Safety and Security Requirements

- SR-1 (Data Encryption): All data transmitted between the mobile application, web portal, and Firebase backend shall be encrypted using Transport Layer Security (TLS 1.3) via HTTPS protocol.
- SR-2 (Authentication Token Expiry): User session tokens generated via UPM SSO and Firebase shall expire after 7 days of inactivity, forcing a re-authentication to secure user access.
- SR-3 (Camera Privacy): The application shall only access the mobile camera during active AR navigation mode. The live camera feed shall be processed locally on-device for tracking and shall not be recorded, stored, or uploaded to any server.
- SR-4 (Access Control): The Firebase database rules shall restrict write permissions strictly to accounts with verified administrative credentials.

5.3 Software Quality Attributes

- QA-1 (Availability): The Firebase cloud backend and administrative web services shall maintain an uptime availability of 99.5% during the academic semester.
- QA-2 (Usability): The user interface shall be intuitive enough that a first-time student or staff member can successfully start a navigation route within 3 screen taps from the main dashboard.
- QA-3 (Reliability): The application shall gracefully handle temporary network dropouts by caching the current indoor routing path locally, allowing the AR guidance to continue based on device sensor dead reckoning until connection is restored.